

Norway

SLEIPNER

Volve F

F-13

F-13

Plan: F-13 Main field rev#5 STGAL 051115

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-13	North Reference:	Grid
Wellbore:	F-13	Survey Calculation Method:	Minimum Curvature
Design:	F-13 Main field rev#5 STGAL 051115	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-13		
Well Position	+N/-S	0.74 m	Northing:
	+E/-W	1.30 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.8315 N
			Longitude:
			1° 53' 15.0081 E
			Water Depth:
			91.00 m

Wellbore	F-13		
Magnetics	Model Name	Sample Date	Declination
			(°)
	MNETICREFERENCE	16.08.2006	-2.45
			Dip Angle
			(°)
			71.61
			Field Strength
			(nT)
			50,330

Design	F-13 Main field rev#5 STGAL 051115		
Audit Notes:			
Version:	Phase:	PLAN	Tie On Depth:
			145.90
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(m)	(m)	(m)
	145.90	0.74	1.30
			Direction
			(°)
			83.35

Survey Tool Program	Date	13.01.2016			
From	To	Survey (Wellbore)	Tool Name	Description	
(m)	(m)				
145.90	220.00	F-13 Main field rev#5 STGAL 051115 (F-13	Inclination Only	Inclination only surveys, vertical wells (inc < 2	
220.00	550.00	F-13 Main field rev#5 STGAL 051115 (F-13	Keeper, sightline	Keeper tool run in sightline mode (initialisation)	
550.00	670.00	F-13 Main field rev#5 STGAL 051115 (F-13	NOT_useMWD gyro, Baker I Baker Hughes' MWD gyro service (GyroTrak)		
670.00	1,449.00	F-13 Main field rev#5 STGAL 051115 (F-13	Magn, IFR, mag-corr, reduce Magnetic Tools (MWD, EMS) without gyro-verificatio		
1,449.00	2,832.00	F-13 Main field rev#5 STGAL 051115 (F-13	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio		
2,832.00	3,433.00	F-13 Main field rev#5 STGAL 051115 (F-13	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio		
3,433.00	5,218.60	F-13 Main field rev#5 STGAL 051115 (F-13	Magn, IFR, mag-corr, dual in- Magnetic Tools (MWD, EMS)		

Planned Survey										
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Vertical	Dogleg	Build	Turn	
Depth	(°)	(°)	Depth	(m)	(m)	Section	Rate	Rate	Rate	
(m)			(m)			(m)	(°/30m)	(°/30m)	(°/30m)	
145.90	0.00	0.00	145.90	0.74	1.30	0.00	0.000	0.000	0.000	
245.90	0.00	0.00	245.90	0.74	1.30	0.00	0.000	0.000	0.000	
365.90	2.00	241.00	365.88	-0.27	-0.53	-1.94	0.500	0.500	0.000	
456.14	5.00	245.00	455.94	-2.70	-5.47	-7.13	1.000	0.997	1.330	
531.14	10.00	245.00	530.27	-6.83	-14.34	-16.42	2.000	2.000	0.000	

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Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-13	North Reference:	Grid
Wellbore:	F-13	Survey Calculation Method:	Minimum Curvature
Design:	F-13 Main field rev#5 STGAL 051115	Database:	Production EDM P246N

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
903.67	30.28	195.67	880.06	-112.59	-69.89	-83.83	2.000	1.633	-3.973	
1,123.62	30.28	195.67	1,070.00	-219.37	-99.85	-125.94	0.000	0.000	0.000	
1,250.30	38.32	191.03	1,174.58	-288.79	-116.02	-150.04	2.000	1.903	-1.098	
2,472.80	38.32	191.03	2,133.76	-1,032.72	-261.08	-380.21	0.000	0.000	0.000	
3,619.10	82.69	43.89	2,948.00	-937.85	247.07	135.50	3.000	1.161	-3.851	
3,988.58	82.69	43.89	2,995.00	-673.75	501.16	418.45	0.000	0.000	0.000	
4,122.16	89.88	48.37	3,003.65	-581.47	597.19	524.50	1.900	1.614	1.005	
4,315.94	89.88	48.37	3,004.06	-452.74	742.02	683.27	0.000	0.000	0.000	
4,446.62	84.03	45.45	3,011.00	-363.65	837.27	788.18	1.500	-1.342	-0.671	
4,812.16	84.03	45.45	3,049.00	-108.58	1,096.34	1,075.03	0.000	0.000	0.000	
5,218.60	67.67	47.68	3,148.00	161.53	1,381.46	1,389.49	1.218	-1.208	0.165	

Design Targets										
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude	
T1 Main Field (elevated - plan hits target center - Rectangle (sides W30.00 H0.00 D16.24)	0.00	0.00	2,948.00	-937.85	247.07	6,477,626.00	435,297.00	58° 25' 59.6325 N	1° 53' 31.1075 E	
T2 Main Field - plan hits target center - Rectangle (sides W0.00 H31.10 D17.69)	0.00	360.00	2,995.00	-673.76	501.15	6,477,890.00	435,551.00	58° 26' 8.3016 N	1° 53' 46.4959 E	
T3 Main Field - plan hits target center - Rectangle (sides W0.00 H38.31 D18.51)	0.00	0.00	3,011.00	-363.65	837.27	6,478,200.00	435,887.00	58° 26' 18.5005 N	1° 54' 6.8950 E	
T4 Main Field - plan misses target center by 0.03m at 4812.16m MD (3049.00 TVD, -108.58 N, 1096.34 E) - Rectangle (sides W0.00 H37.83 D21.76)	0.00	0.00	3,049.00	-108.56	1,096.36	6,478,455.00	436,146.00	58° 26' 26.8800 N	1° 54' 22.6054 E	
T5 Main Field - plan misses target center by 0.19m at 5218.60m MD (3148.01 TVD, 161.67 N, 1381.33 E) - Rectangle (sides W0.00 H50.00 D25.00)	0.00	0.00	3,148.00	161.53	1,381.46	6,478,725.00	436,431.00	58° 26' 35.7576 N	1° 54' 39.9060 E	

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
220.00	220.00	30" Conductor Casing	30.000	36.000	
1,448.73	1,330.27	20" Surface Casing	20.000	26.000	
2,831.95	2,451.89	13 5/8" Production Casing	13.625	17.500	
3,420.00	2,891.63	9 7/8" Production Liner	9.875	12.250	
5,218.60	3,148.00	7" Production Liner	7.000	8.500	

Checked By: _____ Approved By: _____ Date: _____